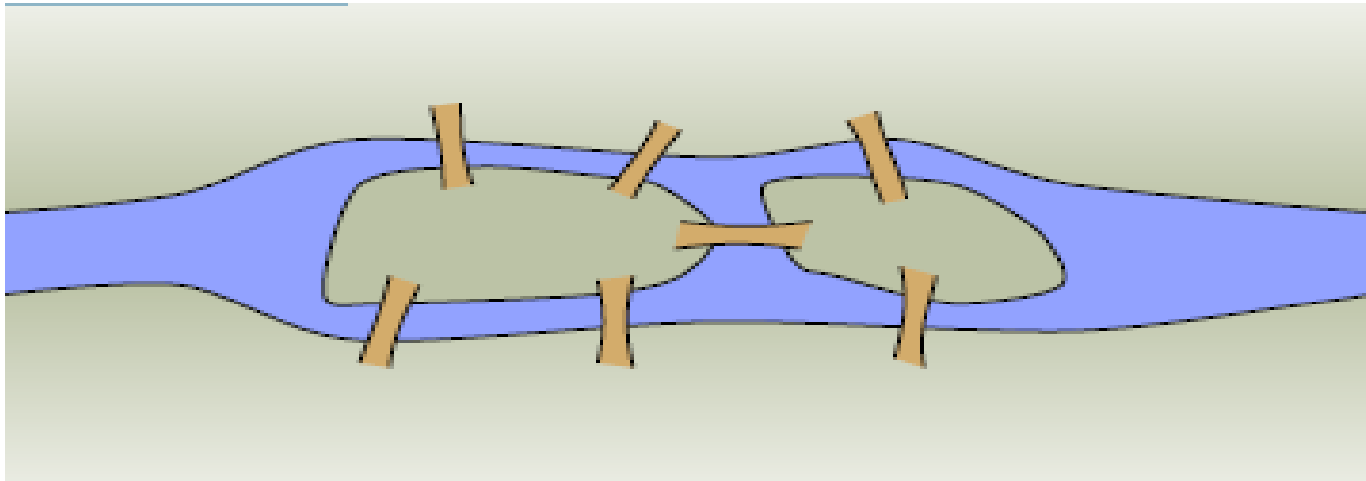


MODULE 5

GRAPH THEORY

In the time of Euler, in the town of Königsberg in Prussia, there was a river containing two islands. The islands were connected to the banks of the river by seven bridges (as seen below). The bridges were very beautiful, so townspeople would spend time walking over the bridges, on their days off



As time passed, a question arose: was it possible to plan a walk so that you cross each bridge once and only once with same starting & ending point?

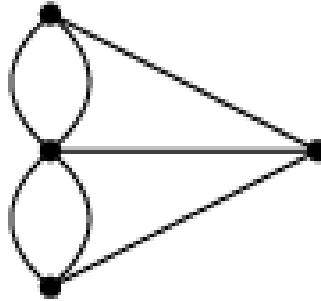
Euler was able to answer this question.

Are you?

The problem above, known as the *Seven Bridges of Königsberg*, is the problem that originally inspired graph theory.

Graph Theory is first studied by the super famous mathematician Leonhard Euler in 1735.

Consider a “different” problem:



Here are four dots connected by some lines.

Is it possible to trace over each line once and only once (without lifting up your pencil, with same starting and ending dot)?

Is there connection between these two problems ?

There is an obvious connection between these two problems

Definition: Graph

A graph is an ordered pair $G = (V, E)$ of two sets V and E satisfying the following conditions:

- (i) V is a finite non empty set.
- (ii) Each $e \in E$ corresponds to a unique **unordered pair** v_1, v_2 of elements of V

The elements of the set V are called **vertices** of the graph (V, E) . A vertex is also referred to as a **node** or a **point**.

The elements of E are called **edges**. An edge is also referred to as a **curve** or a **line**.

Example :

$G=(V=\{a, b, c, d\}, E=\{\{a, b\}, \{a, c\}, \{b, c\}, \{b, d\}, \{c, d\}\})$

Here we have a graph G with $V=\{a, b, c, d\}$ and

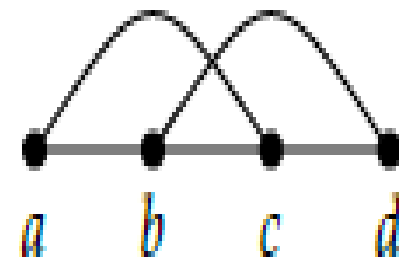
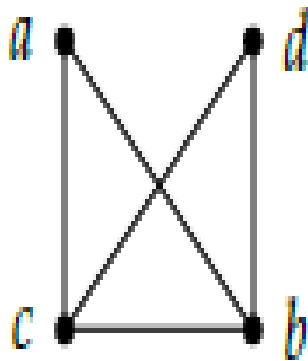
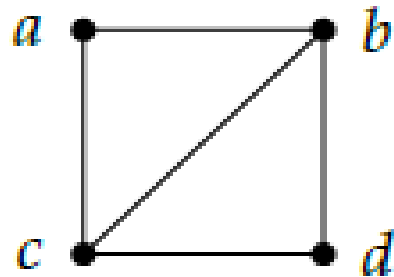
$E=\{\{a, b\}, \{a, c\}, \{b, c\}, \{b, d\}, \{c, d\}\}$

Looking at V, E it is difficult to process.

That is why we often draw a representation of these sets.

We put a dot down for each vertex, and connect two dots with a line precisely when those two vertices are one of the 2-element subsets in our set of edges.

Thus the graph G can be drawn as:



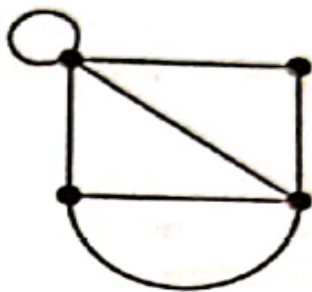
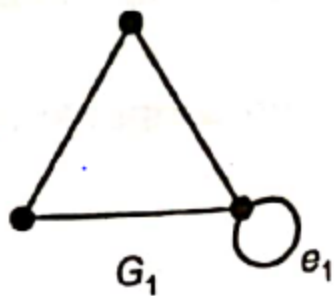
Definition : Loop

If an edge starts and ends at same vertex then it is called as a **Loop**

Definition : Parallel edges/Multiple edges

If two edges have same starting vertex and same end vertex then they are called as **Parallel edges**

Example :



Definitions

Simple graph: A graph having no loops and no parallel edges is called as Simple graph

Degree of a vertex: The number of edges connected to a vertex 'a' is called as degree of a vertex 'a'. It is denoted by $d(a)$.

Null Graph – Graph with No edges

Trivial Graph – Graph with single vertex

Multigraph – Graph with multiple edges

Diagraph – Graph with directed edges

Isolated vertex – Vertex with deg zero

Pendent vertex – Vertex with deg one

Note: for a loop at v, deg is counted twice at v.

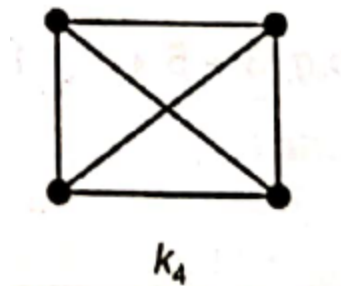
Adjacent Vertices: Two vertices are **adjacent** if they are connected by an edge.

Adjacent Edges: Two edges are **adjacent** if they share a vertex.

Complete graph:

A simple graph in which every pair of vertices is adjacent is called as Complete graph .

The complete graph on n vertices is denoted by K_n



Handshake Lemma

In any graph, the sum of the degrees of vertices in the graph is always twice the number of edges.

$$\sum_{v \in V} d(v) = 2e.$$

Proposition: *In any graph, the number of vertices with odd degree must be even*

In a simple Graph:

Maximum degree of any vertex is $(n-1)$

Why?

Simple graph with n vertices has atmost $n(n-1)/2$ edges

Why?

Definition: If a graph in which degree of every vertex is r then graph is called as **r -regular graph**

2-regular -



3-regular

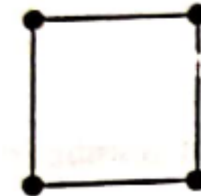


For Given 4 vertices

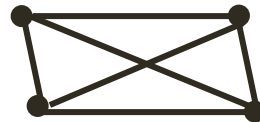
1 regular



2 regular



3 regular



0 regular



Note That,

1. If G is r -regular graph with n vertices and m edges then $m = \frac{nr}{2}$ (Check this for examples)
2. If r and n are odd no. then there can not be an r -regular graph of n vertices. (Check!!)

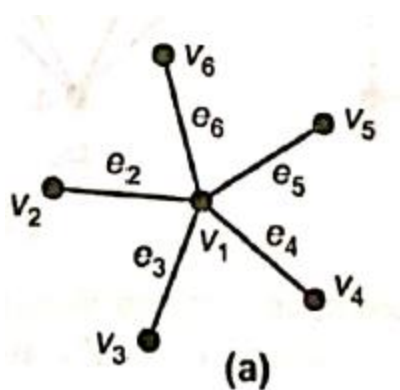
Definition: A graph $G = (V, E)$ is called as **bipartite graph** If

- (i) V can be expressed as union of 2 disjoint sets U, W
- (ii) Every edge in E has one vertex in U and one vertex in W

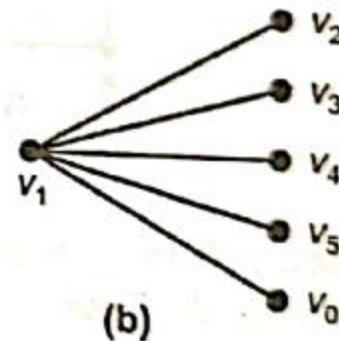
Where U, W are called partition of V

Example :

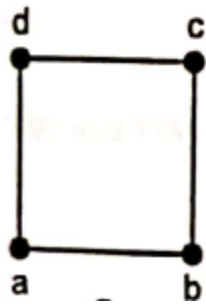
1)



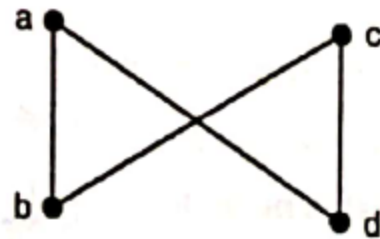
Rearrange



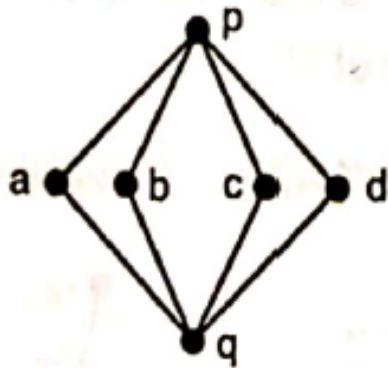
2)



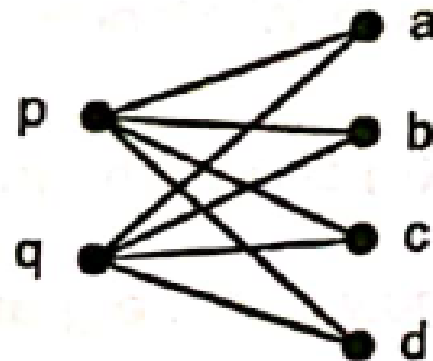
Rearrange



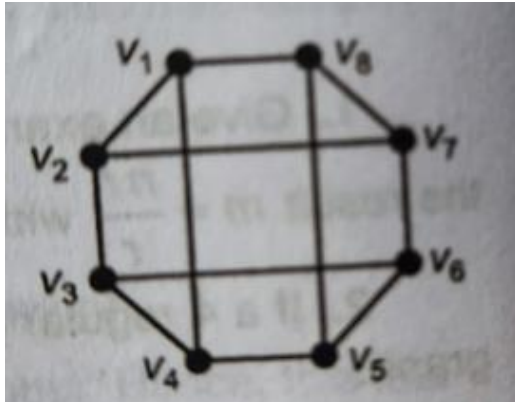
3)



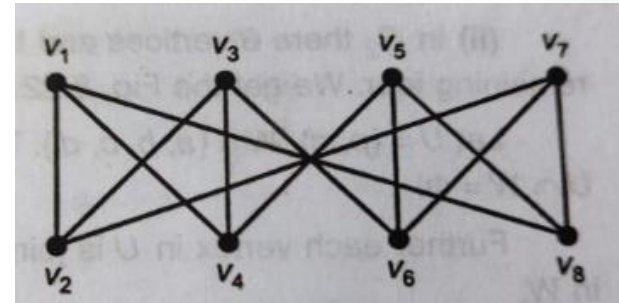
Rearrange



4)



Rearrange

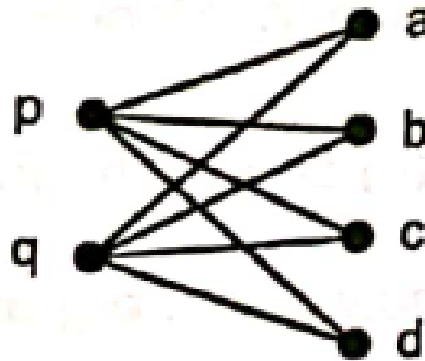


Definition: complete bipartite graph

A bipartite graph $G = (V, E)$ is called as complete bipartite graph if each vertex in U is adjacent to all the vertices in W

complete bipartite graph is denoted by $K_{m,n}$

Where $|U|=m, |W|=n$



Which of the bipartite graphs we have seen are complete bipartite?

Definition : Isomorphic graphs

Let $G1 = (V1, E1)$ & $G2 = (V2, E2)$ be graphs

If there exist a bijective function (isomorphism)

$$f: V1 \rightarrow V2$$

with $\{a, b\} \in E1$ iff $\{f(a), f(b)\} \in E2$

($\{a, b\}$ is an edge of $G1$ iff $\{f(a), f(b)\}$ is an edge of $G2$)

Then $G1$ & $G2$ are called isomorphic graphs

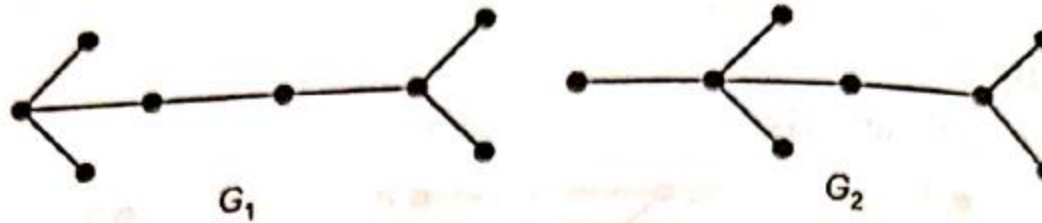
They are denoted by $G1 \cong G2$

Note: For isomorphic graphs, $G1$ and $G2$

1. They must have same number of vertices
2. They must have same number of edges
3. They must have same degree for corresponding vertices. i.e. If $d(a) = 3$ then $d(f(a)) = 3$
4. The isomorphism must preserve the vertex and edge adjacency.

If any one of above conditions is not satisfied then graphs $G1$ and $G2$ are not isomorphic graphs

Example: Check whether graphs below are isomorphic ?



G_1 and G_2 are not isomorphic graphs

In graph G_1 there are 4 vertices of degree 1

In graph G_2 there are 5 vertices of degree 1

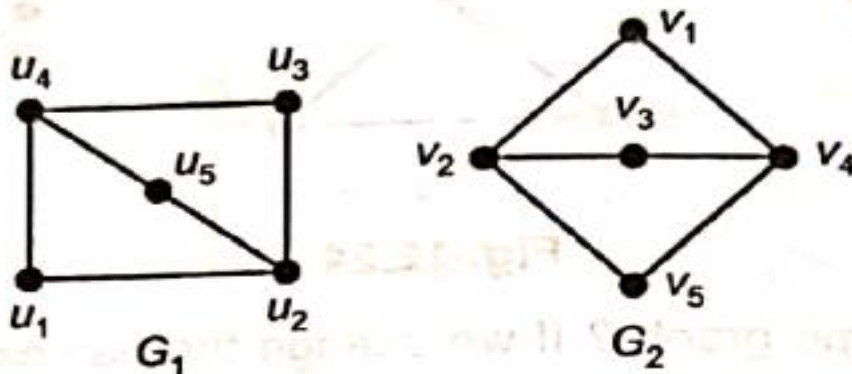
OR

G_2 contains one vertex of deg 4 whereas G_1 does not have any vertex of deg 4

OR

G_1 has 2 vertices of deg 2 whereas G_2 has only one.

Example : Check whether following graphs are isomorphic ?



The two graphs G_1, G_2 have five vertices and six edges each. Further, there are three vertices of degree two and two vertices of degree three.

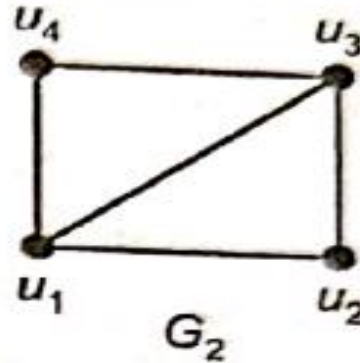
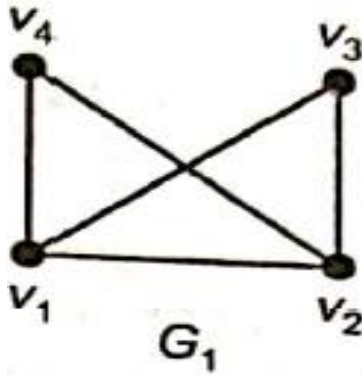
We can define one-to-one correspondence f as follows :

$$u_1 \rightarrow v_5, u_2 \rightarrow v_4, u_3 \rightarrow v_1, u_4 \rightarrow v_2, u_5 \rightarrow v_3.$$

This correspondence preserves the adjacency and incidence relationship

$\therefore$ The graphs are isomorphic.

Example : Check whether following graphs are isomorphic ?



G_1 and G_2 are isomorphic graphs with 4 vertices and 5 edges. There are 2 vertices of deg 2 and 2 vertices of deg 3 in both graphs. One of the isomorphism is given by,

$$v_1 \rightarrow u_1, v_3 \rightarrow u_2, v_2 \rightarrow u_3, v_4 \rightarrow u_4.$$

Check that, it preserves the adjacency.